Shivshakti ProERP is an offline-first UPVC workshop ERP for Shivshakti Enterprises. It covers measurement entry, Sarik-method cutting calculations, internal workshop costing, customer estimates, quotations, labour PDFs, and job document management — on web and Android.

1. Executive Summary

- Single application for sales, workshop, and management workflows
- Centralized calculation engine — one source of truth for cutting, costing, and quotation
- Offline-capable: data stored locally (localStorage + IndexedDB); no server required for core use
- Mobile-ready via Capacitor (Android APK build supported)
- Role-based module access: Admin, Manager, Sales, Workshop

2. Product Types & Calculation Methods

Product	Method / Engine	Key Outputs
UPVC Sliding Window	Sarik Method · Standard · Workshop Confirmed	Sash cuts, glass, interlock, running feet, profile cost, labour PDF
Single Door (Casement)	Sarik Door Method v2.0	Frame, sash, beading, glass, flat mullion (double), door hardware
Double Door (Casement)	Sarik Door Method v2.0	Two-leaf sash, shared flat mullion formula, per-leaf glass/beading
Fixed UPVC Partition Frame	Fixed Frame Engine v1.0	Single frame / separate modules / internal partitions; glass sheet; bar optimization

3. Home Dashboard Modules

Eight role-filtered modules accessible from the Home screen. Smart Estimation Center is the primary entry point for Partner/Manager roles.

Module	Purpose	Typical Roles
Smart Estimation Center	M e a s u r e !’ c a l c u l a t e !’ s a v e !’ q u o t e i r	All except restricted
Cutting Sheet	Labour cutting sizes & technical production details	Admin, Manager, Workshop
Workshop Cost	Internal manufacturing cost, margin, customer pricing	Admin, Manager
Estimate	Multi-product preliminary customer estimates	Admin, Manager, Sales
Quotation	Create, revise, share customer quotations	Admin, Manager, Sales
Documents	PDF history, revisions, downloads	All roles
Profiles & Rates	Profile catalog, kg rates, hardware master	Admin, Manager

Module	Purpose	Typical Roles
Settings	Company letterhead, taxes, diagnostics	Admin, Manager

4. Smart Estimation Center

- Product selector: Window · Single Door · Double Door · Fixed Frame
- Measurement entry in mm or ft/in with live validation
- Cutting summary, profile cost, workshop cost, and product preview
- Save product to job, add to quotation, export labour cutting PDF
- AI Assistant: text and voice commands for calculate, navigate, and read results
- Add to Estimate for multi-opening project estimates

5. Job & Workflow Management

- Structured job IDs (JOB-YYYY-NNNN) with multi-product support (W-001, D-001, F-001)
- Job statuses: draft → measured → estimation-ready → quotation-sent → approved →
- Activity audit trail per job (product edits, documents, quotation revisions)
- Quotation revision workflow with superseded/approved states
- Document registry with blob storage (IndexedDB) and revision numbering

6. Documents & PDF Outputs

- Labour Cutting PDF — component sizes, cut tables, optional preview diagrams
- Quotation PDF — Shivshakti letterhead, GST modes, grouped line items
- Estimation / proposal PDFs (premium proposal module)
- Summary profile weight PDF
- Company settings flow into quotation headers and footers dynamically

7. Fixed Frame Module (Latest)

- Construction modes: SINGLE_FRAME · SEPARATE_MODULES · INTERNAL_PARTITIONS
- Division modes: NONE · AUTO · EQUAL · MANUAL with joint gaps and clearances
- Profile-specific glass deductions (frame + bead + thickness)
- Reinforcement rules: full perimeter, horizontal, vertical, custom
- Bar optimization (5800 mm standard, kerf, reusable leftover tracking)
- Workshop cost: profiles, glass, hardware/consumables, labour, transportation
- No sliding track, sash, roller, handle, lock, or mesh logic

8. User Roles & Permissions

Role	Access Highlights
Admin	Full access; company settings; finalize documents; legacy Summary tab
Manager	Smart Estimation-first; jobs, documents, quotations; internal rates
Sales	Customer quotations and estimates; no internal workshop rates
Workshop	Labour sheets and cutting; no internal costing or sales tools

9. Technical Architecture

- Frontend: Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS
- Mobile: Capacitor 7 — Android APK (debug/release build scripts included)
- PDF: jsPDF + jspdf-autotable; pdf-lib for document assembly
- Storage: localStorage (jobs, settings, counters); IndexedDB (PDF blobs)
- Calculation libs: lib/calculations, lib/cutting, lib/estimation, lib/summary
- Verification: 10+ automated scripts (verify:all) for cutting, doors, fixed frame, consistency

10. Verification & Quality Assurance

- verify:cutting — Sarik window acceptance samples
- verify:door-cutting — Sarik door v2 sash height (+3 mm correction)
- verify:fixed-frame — partition frame, Ahilyanagar 385 sq.ft regression
- verify:calculations — end-to-end product ⁉ summary ⁉ quotation
- verify:consistency — cross-module parity for all product types
- verify:workflow — job IDs, quotation revisions, product lifecycle
- verify:assistant — AI command parser regression

11. Known Limitations & Roadmap Notes

- Roles are device-local (localStorage) — not server-enforced multi-user auth
- Optional OpenAI assistant API unavailable in static APK export (local parser used offline)
- Fixed frame manual module drag-resize UI not yet implemented
- Email API route optional — WhatsApp/share uses local PDF when server unconfigured

Contact & Product Information

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Application	Shivshakti ProERP | UPVC Workshop
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